❖ Gymkhana Management System:

Introduction

The **Gymkhana Management System** is a digital platform developed to manage all sports and fitness-related activities within a college.
It automates the daily operations of the gymkhana — including **member registration, sports event scheduling, equipment inventory, attendance tracking, and performance reports.**
This system encourages **active participation, transparency, and efficient coordination** among students, coaches, and the administration.

Objectives

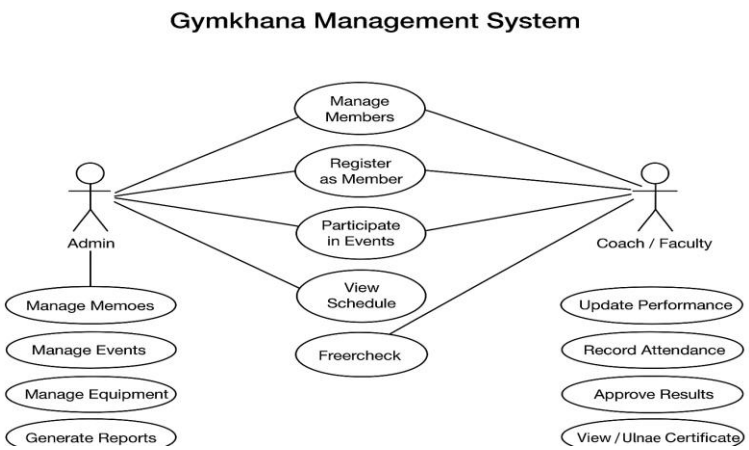
- To maintain **digital records** of gymkhana members and sports inventory.
- To **organize and manage** inter-department and inter-college sports events.
- To **track student participation, achievements, and awards.**
- To ensure **equipment issue/return** and maintenance are handled efficiently.
- To **generate analytical reports** for sports performance and participation statistics.

Modules / Functional Areas

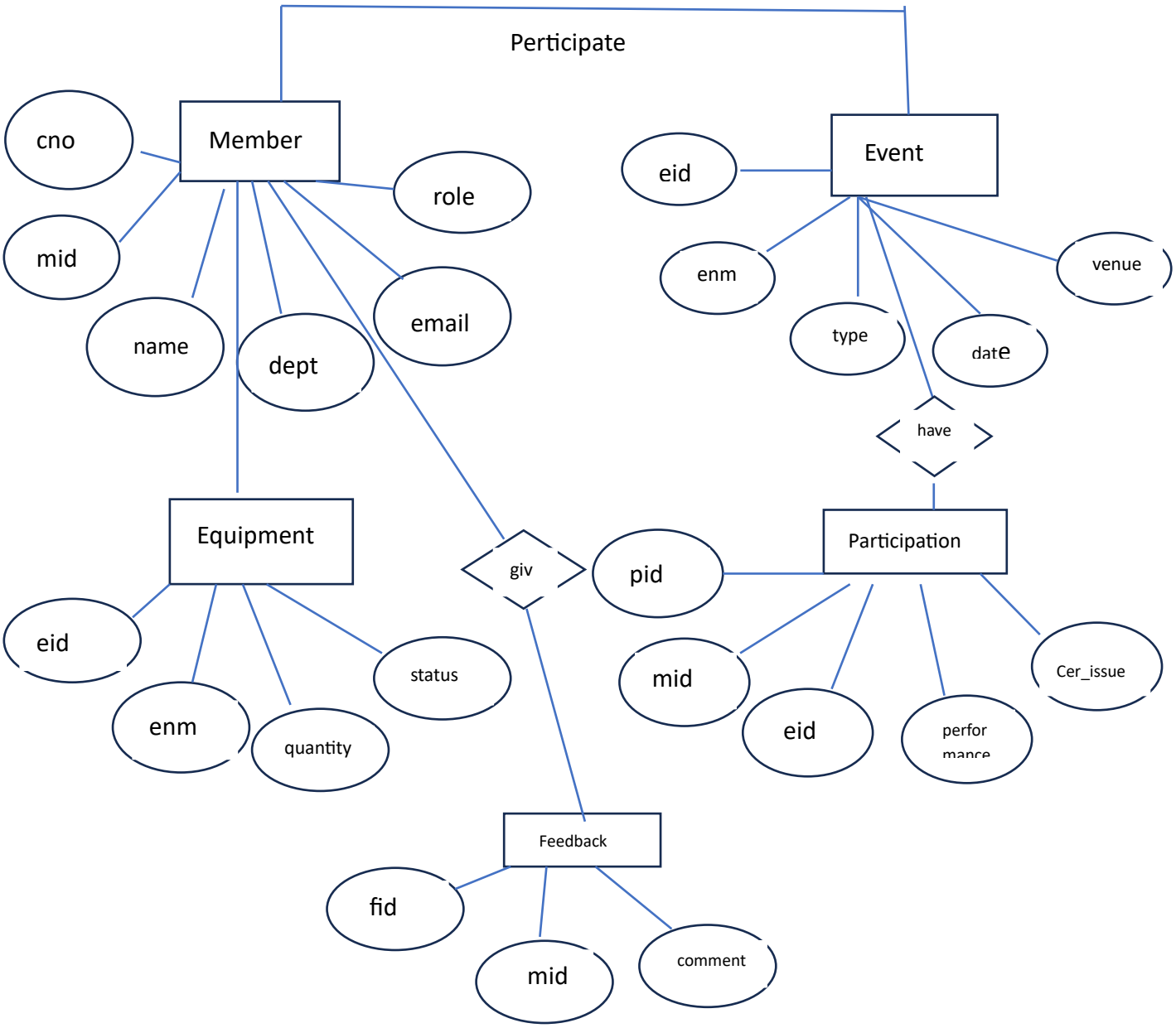
1. **Member Registration Module**
    - Students and staff can register as gymkhana members.
    - Membership records include name, department, sport preference, and validity.
    - Allows easy renewal and ID-based tracking.
2. **Event Management Module**
    - Schedule and manage sports tournaments, matches, and fitness programs.

- Notify participants through digital notice boards or email alerts.
- Record match results, winners, and points.

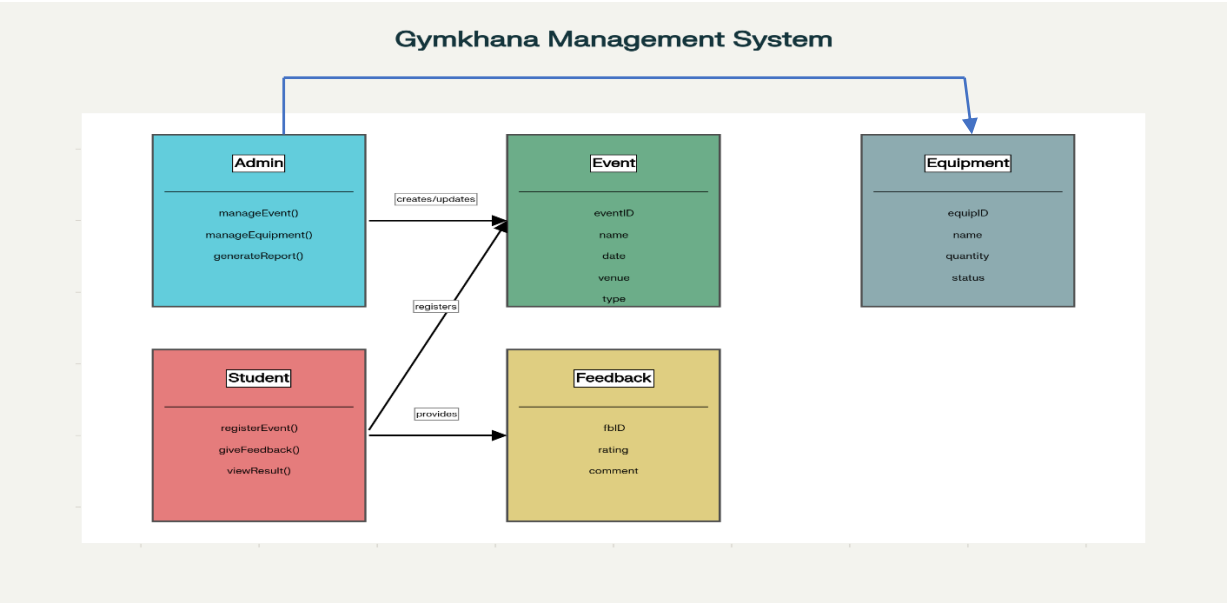
3. **Equipment Management Module**
   - Maintain an inventory of all sports items (bats, rackets, balls, etc.).
   - Track issue/return dates and send maintenance alerts.
   - Reduce manual loss or damage reporting.
4. **Performance and Attendance Tracking**
   - Track student participation in each sport or event.
   - Record achievements, medals, and certificates.
   - Automatically calculate attendance for sports practices and fitness sessions.
5. **Reports and Certificates**
   - Generate event summary reports, participation certificates, and performance charts.
   - Useful for **NAAC/NBA accreditation** and internal sports evaluation



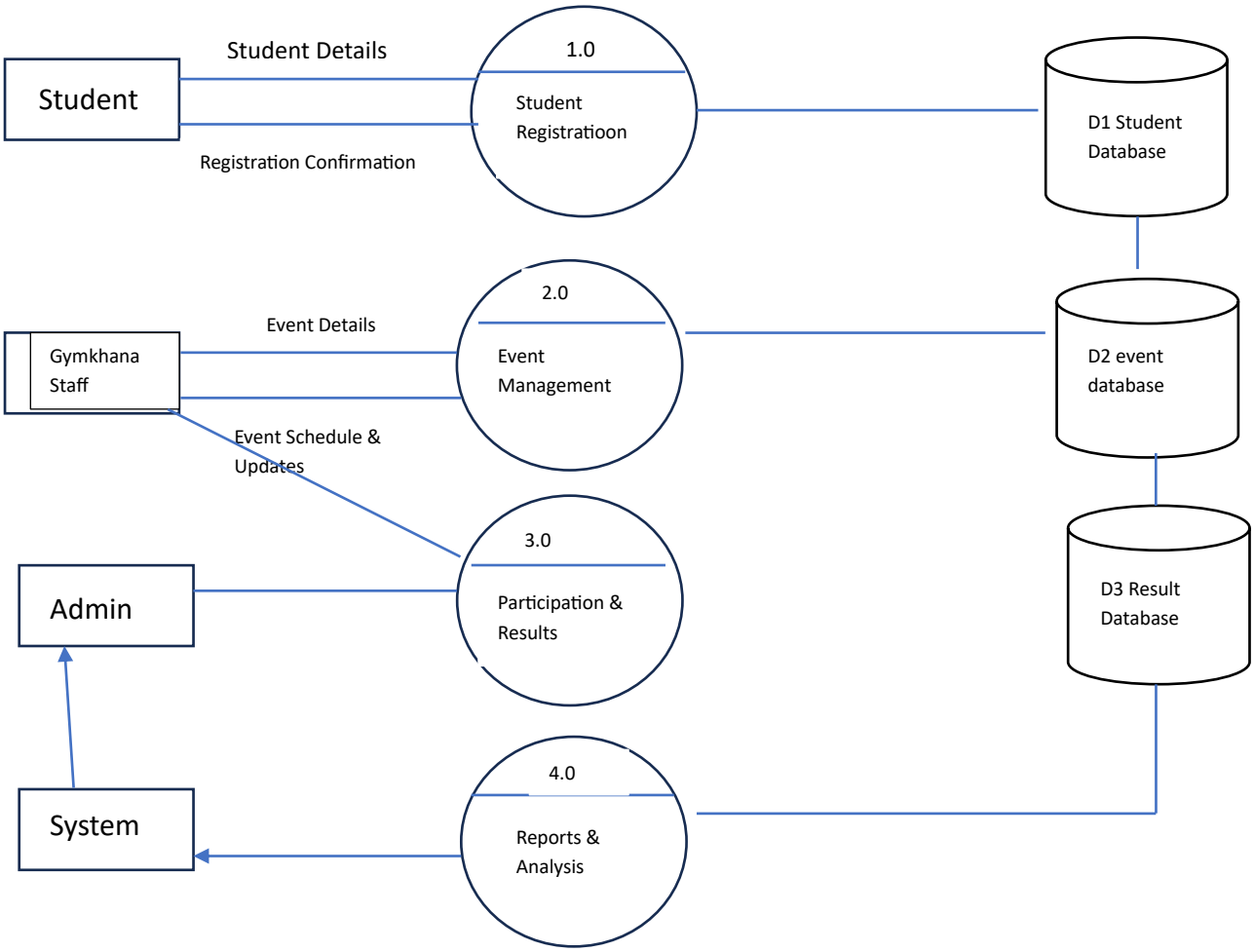
Use Case for Gymkhana Management System



ER diagram for Gymkhana management



UML diagram for gymkhana management



DFD diagram